

The Second Logarithmic Term for Regular Tesler Matrices

A uniform entropy construction and integer throughput decomposition

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Prepared for Vladimir

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Abstract

Let a_n count the $n \times n$ nonnegative integer Tesler matrices with all hook sums equal to one, as in OEIS A008608. We prove

$$\log a_n = Cn^{3/2} - \frac{3}{4}n \log n + O(n \log \log n), \quad C = \pi \sqrt{\frac{8}{27}}.$$

The leading logarithmic equivalent is an established result of Balashov, Bulavenko and Molybog (2026). The refinement here is the coefficient $-3/4$ of $n \log n$. An elementary partition-product upper bound is matched at this scale by a feasible geometric-entropy flow, independent triangle moves producing many integer throughput vectors, and the existing arbitrary-support contingency-table capacity bound of Brändén, Leake and Pak. The proof controls the initial and terminal boundaries and is uniform as the number of variables grows.

We give a two-sided refinement with lower loss $2n \log \log n + O(n)$, an explicit linear constant in the upper estimate, and the resulting next correction to the integer threshold inverse. Exact rational verification of finite combinatorial identities is supplied separately from floating-point sanity checks. An $O(n)$ remainder, multiplicative equivalent, amplitude, and full expansion remain unproved.

[write] Status and trust boundary, added 5 October 2026, batch 103. This is bundle Report 106, filed in ProveIt as a single-source research report (provenance, credits, notation and the collected non-claims are in Section 1.1). It is AI-assisted, unrefereed and not formalized. Its title page reads “Prepared for Vladimir” (Vladimir Reshetnikov, who maintains the repository), as delivered. The leading term $\log a_n \sim Cn^{3/2}$, and with it the leading inverse $N(y) \sim (\log y/C)^{2/3}$, is due to Balashov, Bulavenko and Molybog [2, Corollary 37]. The counting inequality of the lower bound is Theorem 2.1 of Brändén, Leake and Pak [3], used as an external input. New here are the coefficient $-3/4$ of $n \log n$ with the explicit two-sided bounds of Theorem 1.1, and the second term of the integer inverse with an integer bracket (Theorem 10.1). Whether the coefficient $-3/4$ is new to the literature is *not* certified (Question 6). All O -constants are non-effective, and the numerical checks are not interval certificates.

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1 Statement and relation to prior work

All logarithms are natural. We use the matrix-size indexing $a_0 = 1$ and

$$a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8 = 1, 2, 7, 40, 357, 4820, 96030, 2766572.$$

The OEIS entry starts at $n = 1$ [1]. Define

$$c = \frac{\pi}{\sqrt{6}}, \quad C = \frac{4c}{3} = \pi\sqrt{\frac{8}{27}}, \quad B_U = \frac{3}{4} + \frac{3}{2}\log c - \frac{1}{2}\log(2\pi). \quad (1.1)$$

Numerically, $B_U = 0.204336693648386\dots$; this decimal is descriptive and is not used in a proof.

Theorem 1.1. *For regular Tesler matrices, as $n \rightarrow \infty$,*

$$\log a_n \geq Cn^{3/2} - \frac{3}{4}n \log n - 2n \log \log n - O(n), \quad (1.2)$$

$$\log a_n \leq Cn^{3/2} - \frac{3}{4}n \log n + B_U n + O(\sqrt{n}). \quad (1.3)$$

In particular,

$$\log a_n = Cn^{3/2} - \frac{3}{4}n \log n + O(n \log \log n), \quad \lim_{n \rightarrow \infty} \frac{Cn^{3/2} - \log a_n}{n \log n} = \frac{3}{4}. \quad (1.4)$$

The constants implicit in $O(\cdot)$ are absolute and independent of n .

What is already known. Balashov–Bulavenko–Molybog [2, Corollary 37] prove $\log a_n \sim Cn^{3/2}$. Their paper was published on 8 May 2026. That leading result and its leading inverse consequence are not claimed as new here. Brändén–Leake–Pak [3, Theorem 2.1] supply the counting inequality used in the lower proof. Leake–Morales [4] develop the corresponding capacity theory for integral flows and the Kostant partition function; their work provides important context, while the proof below uses the arbitrary-cell-bound theorem of [3] directly.

Scope of the refinement. The theorem determines a logarithmic coefficient. Exponentiating an $O(n \log \log n)$ error does not produce a relative-error asymptotic for a_n . In particular B_U is a constant in an upper bound, not a proved coefficient of n in $\log a_n$. A bounded source check through 2 October 2026 did not locate the $-3/4$ refinement in the primary works consulted. This does not certify exhaustive novelty, and no priority claim is needed for the proof.

Proof architecture. Singleton elimination makes the upper bound inexpensive. For the lower bound, a geometric profile has entropy $Cn^{3/2} - \frac{1}{4}\log(n!) - O(n)$. Each constructed bulk throughput is only $O(\sqrt{j} \log(j+1))$, so the two margin penalties together cost at most $\log(n!) + 2n \log \log(n+1) + O(n)$. Independent local moves supply $\exp(\frac{1}{2}\log(n!) - O(n))$ disjoint integer margin vectors. The three factorial coefficients therefore combine as $-\frac{1}{4} - 1 + \frac{1}{2} = -\frac{3}{4}$.

1.1 Provenance, credits and reading conventions

[write] **Added 5 October 2026, batch 103.** *Provenance.* This report is Report 106 of the session bundle that ProveIt’s docs/incoming intake processed as batch 103: the archive A008608_Tesler_Second_Logarithmic_Term_Source.zip (402,353 bytes, wrapper directory report106/, 22 files), arrival commit 60f54ea06, placed in 9c995cefe. The text is the delivered report106.tex, shipped as article.tex. The delivered 13-page PDF report106.pdf, the checksum manifest manifest.json and its verifier verify_package.py are not shipped; all three can be retrieved from the arrival

commit. The package names no ProveIt commit, path or report, so the report has no pin. Its title page and PDF metadata say that it was prepared for Vladimir; it carries no “prepared for private review” line. It is a single-source report, so no merge choice was made. The write prefixed the 78 delivered labels with `tes:` (“Tesler”), and added the notes marked [write], this subsection, Remarks 4.1 and 9.2, the references [6, 7, 8, 9], and Section 12.1. No statement, proof or number of the source was changed. Theorem and equation numbers follow the section counter (Theorem 1.1 is Theorem 1.1, and so on).

Inherited material.

- *The leading term* $\log a_n \sim Cn^{3/2}$ is Corollary 37 of Balashov, Bulavenko and Molybog [2] (p. 17 of the journal version, published 8 May 2026; their notation is $\mathcal{L}(1^n) = \ln T(1^n)$). The manuscript credits it, together with the leading inverse $N(y) \sim x$ of Theorem 10.1, which follows from it. The intake fetched the journal PDF, confirmed that its SHA-256 equals the fingerprint in the delivered `data/sources.json`, and checked that the constant is $\pi\sqrt{8/27} = C$. Balashov, Bulavenko and Molybog describe their result as a drastic improvement of Leake and Morales [4, Theorem 1.2].
- *The counting inequality* of the lower bound is Theorem 2.1 of Brändén, Leake and Pak [3], restated as Theorem 8.1; the note after that theorem records the intake’s comparison with the source.
- *Classical inputs:* the Dedekind eta transformation [5, (23.18.5)] behind (3.7); the identification of regular Tesler matrices with integral flows of netflow $(1, \dots, 1, -n)$, that is, with the type A Kostant partition function at that vector (OEIS [1]; see also [7]). The first-row recursion (2.4) is elementary and proved in the text.

Earlier bounds, not cited by the manuscript. Mészáros, Morales and Rhoades [7, §4] gave $n! \leq a_n \leq 2^{\binom{n}{2}}$ (as O’Neill reports them). O’Neill [6] raised the lower bound to $(2n-3)!! = \prod_{i=1}^{n-1} (2i-1)$ and tightened the upper bound, which stays of the form $e^{O(n^2)}$: Theorem 43 of the journal version (Theorem 5.6 of arXiv:1702.00866v2; the two statements agree). He also records a question of Pak, whether $a_n = e^{\Theta(n^2)}$, as Question 4 of the journal version (Question 1.4 of the arXiv version); Balashov, Bulavenko and Molybog recall it, without a number, in their Section 2. In Remark 5 of the journal version (an unnumbered remark after Question 1.4 in the arXiv version) he adds that OEIS data suggest $\log a_n = O(n^{1.6})$, an observation he attributes to J. Pantone (a private communication to A. H. Morales, 2016). (Corrected 5 October 2026 after the independent check recorded at the end of Section 12.1, which read both versions; from “ $e^{O(n^2)}$ ” to here the text first read “(Theorem 5.6 of the arXiv version 1702.00866v2, which the intake read; the journal version was not compared). O’Neill also records a question of Pak, whether $a_n = e^{\Theta(n^2)}$ (his Question 1.4, which Balashov, Bulavenko and Molybog recall in their Section 2), and an observation that OEIS data suggest $\log a_n = O(n^{1.6})$.”) Corollary 37 of [2] answers Pak’s question in the negative; the elementary upper bound (4.2) of this report gives the same answer on its own, since $U_n = O(n^{3/2})$. Indeed F is decreasing, so $L(t) \leq \int_0^\infty F(xt) dx = \pi^2/(6t)$; with $\log(1 - e^{-t}) < 0$, $c^2 = \pi^2/6$ and $\sqrt{j} \leq \int_j^{j+1} \sqrt{x} dx$ this gives, for every $n \geq 1$ and without the eta transformation,

$$\log a_n \leq U_n \leq 2c \sum_{j=1}^n \sqrt{j} \leq C(n+1)^{3/2}.$$

In §7.3 of [4] (arXiv v3), finished before its authors learned of the preprint of [2] (their footnote 2), $O(n^2)$ is still named as the best known upper bound on $\log a_n$. (The last two sentences were added 5 October 2026 after the same check.) None of these bounds reaches the scale of the second term, so the omission of [6] affects no proof here; it bears only on the priority question below. The wider Tesler literature listed on the OEIS page (Haglund [9]; Armstrong, Garsia, Haglund, Rhoades and Sagan [8]; [7]) concerns diagonal harmonics, the Tesler polytope and weighted sums (judging by their titles); the intake did not search it for asymptotics.

What is new, and its status. New relative to the papers above and to the repository: the coefficient $-3/4$ of $n \log n$ with the two-sided bounds (1.2)–(1.3), and the second term $(2C)^{-1} \sqrt{x} \log x$ of the integer inverse with the bracket (10.7). The source’s own literature check was bounded (“Scope of the refinement” above). The “Future questions” of [2, §5] do not ask for the second term, so the manuscript answers no named question. The intake did not find the coefficient $-3/4$ in [2], [3], either version of [6] or the OEIS page (which has no asymptotic formula), and [4] works at a coarser scale: its Tesler bound, Theorem 1.2 of arXiv v3, is the lower bound $\log a_n \geq \frac{n}{4} \log^2 n - O(n \log n)$, and its §7.3 is quoted above. No exhaustive search was made. (Updated 5 October 2026 after the independent check; these two sentences first read “The intake found the coefficient $-3/4$ neither in [2], [3] or the arXiv version of [6],

nor on the OEIS page (which has no asymptotic formula); it did not compare [4, Theorem 1.2] in detail and made no exhaustive search.”) Novelty is therefore *uncertified* (Question 6).

Repository. Before batch 103 no file of ProveIt mentioned Tesler matrices, A008608, Lusztig data, the Kostant partition function or flow polytopes. The word “contingency” occurs only in **a260700-parabolic-double-cosets** (tables counted by A120733), which shares no theorem with this report. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status.

Indexing. The OEIS entry A008608 has offset 1 ($a_1 = 1, a_2 = 2, a_3 = 7, \dots$); the manuscript adds $a_0 = 1$ (the empty array), which is a convention, not an OEIS term.

Notation. The manuscript reuses several letters. The table fixes each meaning; no symbol was renamed.

Symbol	Meanings (where)	Not to be read as
c, C	$c = \pi/\sqrt{6}$ and $C = 4c/3 = \pi\sqrt{8/27}$ (1.1)	
B_U, B_n	B_U the constant of the upper bound (1.1), (1.3); B_n the width of the terminal layer (6.3)	B_U as a coefficient of n in $\log a_n$ (Remark 4.1); B_n as B_U
λ	$\lambda_i = c/\sqrt{i}$ (Section 4), $\lambda = c/\sqrt{i}$ (proof of Lemma 6.1), any $\lambda > 0$ in (3.9); the row sums $\lambda_i = \sum_j k_{ij}$ of the cell bounds in Theorem 8.1	the λ_i of Theorem 8.1 as $c/\sqrt{i}$
K, k	K absolute constants (proofs of Lemmas 6.1, 6.2, (6.4), Proposition 6.3); the cell-bound matrix $K = (k_{ij})$ in CT_K, P_K (Section 8)	
h	the entropy function $h(u)$ (3.2); the hook-sum vector $h = (h_1, \dots, h_n)$ in (2.4)	
T, S	$T(h)$ the number of matrices with hook sums h (2.4); T_i the cut loads of the profile (5.2); T_i^∞ those of the untruncated profile (Lemma 6.1); $S_i(m)$ the cut loads of an interval array (2.2)	T_i as a count
L	$L(t) = \sum_\ell F(\ell t)$ (3.1); $L = \log y$ (Section 10)	
G	a feasible real table G (Lemma 8.2); $G(t) = Ct^{3/2} - \frac{3}{4}t \log t$ (proof of Theorem 10.1)	
m, M	$m_{[a,b]}$ an interval array (2.2); m the number of table rows in Theorem 8.1; $m_\pm$ the bracket points (proof of Theorem 10.1); $M = 1024$ the cutoff (5.1)	M as a row count
u, v	$u \geq 0$ the argument of h and of (8.5); u_i, v_j the truncated margins of Theorem 8.1	
r, ρ, R	r_j the real outgoing total at vertex j (5.6); ρ_j its integer replacement; $\mathcal{R}_n$ the set of vectors ρ (7.3); $R(t)$ (3.1); $\mathbb{R}$ the reals	
x, y, z	formal variables x_i (2.3); the variables x_i, y_j of P_K ; $y > 1$ the threshold and $x = (L/C)^{2/3}$ (Section 10); z in (8.5), and $z = x + (2C)^{-1}\sqrt{x} \log x$ (proof of Theorem 10.1)	the x of Section 10 as a formal variable
t, s	$t > 0$ the argument of Q, F, L, R, E ; $t_j \in [0, w_j/2]$ the triangle amounts (7.2); $s \geq 4\pi^2/c$ (proof of Lemma 3.1) and $s = \sqrt{x} \log \log x$ (proof of Theorem 10.1)	t_j as an argument t
D, N	D_n the terminal entropy loss (proof of Proposition 5.1); D the bracket constant (10.7); $N(y)$ the integer threshold (10.1)	
f	$f_{[a,b]}$ indexed by intervals; in (7.2) $f_{(j-1,j+1)}$ is indexed by the edge $(j-1, j+1)$, which is the interval $[j, j+1]$	an edge subscript as an interval
subscripts	capacity (Section 8); the preamble redefines the <code>amssymb</code> command of that name (a double-intersection symbol), which the text never uses	

What is not claimed (collected from the source): the leading equivalent and its leading inverse are not new; no $O(n)$ remainder, multiplicative equivalent, amplitude, full expansion or exhaustive novelty certification is claimed; exponentiating an $O(n \log \log n)$ error does not give a relative-error asymptotic, and B_U is a constant in an upper bound, not a proved coefficient of n in $\log a_n$; the bounded source check through 2 October 2026 does not certify novelty, and no priority claim is needed for the proof; the inverse error is asymptotic with an unspecified constant, not a certified numerical error bar or an $O(1)$ inverse; the proof invokes no fixed-dimensional saddle theorem in a growing dimension and no unproved local central limit approximation; the exact checks do not certify a uniform asymptotic estimate or replace any lemma, and their fixtures are manufactured, not the profile (5.2); the floating sanity checks are not interval arithmetic and prove no analytic inequality; the snapshot hashes are regression snapshots, not authentication, and the corruption tests are no protection against simultaneous malicious edits; external scholarly PDFs are fingerprinted, not redistributed. Open questions are collected in Section 12.1.

2 Exact matrix interval and flow models

A regular Tesler matrix is an upper-triangular matrix $A = (A_{ij})_{1 \leq i, j \leq n}$ with entries in $\mathbb{Z}_{\geq 0}$ and hook sums

$$A_{ii} + \sum_{k=i+1}^n A_{ik} - \sum_{k=1}^{i-1} A_{ki} = 1 \quad (1 \leq i \leq n). \quad (2.1)$$

Associate A_{ij} for $i < j$ with the edge $(i-1, j-1)$, and A_{ii} with $(i-1, n)$, on the complete directed acyclic graph with vertices $0, 1, \dots, n$. Then (2.1) says that the net outflow is 1 at each vertex $0, \dots, n-1$, and $-n$ at vertex n .

For every interval $[a, b]$, $1 \leq a \leq b \leq n$, let $m_{[a,b]}$ be the flow on the edge $(a-1, b)$. Its cut loads are

$$S_i(m) = \sum_{a \leq i \leq b} m_{[a,b]}, \quad 1 \leq i \leq n. \quad (2.2)$$

Summing netflows to the left of the cut gives $S_i = i$. Conversely, differences of successive cut loads give all netflows, with the two endpoint conditions included. Thus

$$a_n = [x_1 x_2^2 \cdots x_n^n] \prod_{1 \leq a \leq b \leq n} (1 - x_a x_{a+1} \cdots x_b)^{-1}. \quad (2.3)$$

The staircase $1, 2, \dots, n$ consists of *cut loads*; the netflow is the constant vector $(1, \dots, 1, -n)$. A staircase netflow would be a different counting problem.

Lemma 2.1 (Singleton elimination). *The same a_n counts the nonnegative integer arrays on intervals $a < b$ with nonsingleton cut loads at most i at cut i .*

Proof. Given the nonsingletons, set

$$m_{[i,i]} = i - \sum_{\substack{a \leq i \leq b \\ a < b}} m_{[a,b]}.$$

These are the unique possible singleton values. They are nonnegative integers exactly under the stated inequalities. $\square$

An independent exact recurrence follows by removing the first row of a Tesler matrix. If $T(h_1, \dots, h_n)$ counts matrices of nonnegative integer hook-sum vector h , then

$$T(h_1, \dots, h_n) = \sum_{\substack{x_2, \dots, x_n \geq 0 \\ x_2 + \dots + x_n \leq h_1}} T(h_2 + x_2, \dots, h_n + x_n), \quad T(\emptyset) = 1. \quad (2.4)$$

The omitted diagonal entry is $h_1 - \sum x_j$; hence each summand has multiplicity one. In particular $a_n = T(1, \dots, 1)$. The supplied finite checker compares this recurrence with interval enumeration.

3 One dimensional partition and entropy estimates

Put, for $t > 0$ and $u \geq 0$,

$$Q(t) = \frac{1}{e^t - 1}, \quad F(t) = -\log(1 - e^{-t}), \quad L(t) = \sum_{\ell \geq 1} F(\ell t), \quad R(t) = \sum_{\ell \geq 1} \ell Q(\ell t), \quad (3.1)$$

$$h(u) = (u + 1)\log(u + 1) - u \log u, \quad h(0) = 0, \quad E(t) = h(Q(t)) = tQ(t) + F(t). \quad (3.2)$$

All these series converge absolutely at every positive argument. The function h is increasing, with

$$h'(u) = \log(1 + u^{-1}) \leq u^{-1} \quad (u > 0), \quad \log(1 + u) \leq h(u) \leq \log(1 + u) + 1. \quad (3.3)$$

The upper inequality follows from $u \log(1 + 1/u) \leq 1$; the value at zero is by continuity.

Lemma 3.1 (Partition estimates). *Uniformly for $0 < t \leq c$,*

$$L(t) = \frac{\pi^2}{6t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{t}{24} + O(e^{-4\pi^2/t}), \quad (3.4)$$

$$R(t) = \frac{\pi^2}{6t^2} - \frac{1}{2t} + O(1), \quad (3.5)$$

$$\sum_{\ell \geq 1} E(\ell t) = \frac{\pi^2}{3t} + \frac{1}{2} \log \frac{t}{2\pi} + O(1). \quad (3.6)$$

Also $R(t) \leq \pi^2/(6t^2)$ for every $t > 0$.

Proof. The classical Dedekind eta transformation [5, §23.18], applied on the positive imaginary axis, gives the exact identity

$$L(t) = \frac{\pi^2}{6t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{t}{24} + L(4\pi^2/t). \quad (3.7)$$

Indeed, substitute $\eta(it/(2\pi)) = e^{-t/24} \prod_{\ell \geq 1} (1 - e^{-\ell t})$ into $\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau)$ and take logarithms. For $s \geq 4\pi^2/c$, the series gives $L(s) = O(e^{-s})$, proving (3.4). Differentiating the exact identity, rather than an unspecified error term, gives

$$R(t) = \frac{\pi^2}{6t^2} - \frac{1}{2t} + \frac{1}{24} - \frac{4\pi^2}{t^2} R(4\pi^2/t).$$

The last term is bounded uniformly on $(0, c]$, since $R(s) = O(e^{-s})$ for large s . This proves (3.5). Since $\sum E(\ell t) = tR(t) + L(t)$, (3.6) follows.

Finally $x/(e^{tx} - 1)$ is decreasing for $x > 0$. Its right Riemann sum is no greater than its integral, so

$$R(t) \leq \int_0^\infty \frac{x}{e^{tx} - 1} dx = t^{-2} \sum_{k \geq 1} \int_0^\infty x e^{-kx} dx = \frac{\pi^2}{6t^2}.$$

Positive summands justify the interchange. □

Lemma 3.2 (Entropy tails). *The function E is positive and decreasing, and*

$$\int_0^\infty E(x) dx = \frac{\pi^2}{3}, \quad \int_0^\infty x E(x) dx = 3\zeta(3). \quad (3.8)$$

For every $\lambda > 0$,

$$\sum_{\ell \geq 1} \ell E(\ell \lambda) \leq \frac{3\zeta(3)}{\lambda^2} + \frac{\pi^2}{3\lambda}. \quad (3.9)$$

Proof. Differentiation gives $E'(t) = -te^t/(e^t - 1)^2 < 0$. Near zero $E(t) = O(1 + |\log t|)$; at infinity $E(t) = O((1 + t)e^{-t})$. Expanding $Q(x) = \sum_{k \geq 1} e^{-kx}$ and $F(x) = \sum_{k \geq 1} e^{-kx}/k$ proves (3.8) by termwise integration. For $x \in [(\ell - 1)\lambda, \ell\lambda]$, $\ell\lambda \leq x + \lambda$ and $E(\ell\lambda) \leq E(x)$. Integrate their product and sum to obtain (3.9). □

4 The elementary upper bound

Set $\lambda_i = c/\sqrt{i}$. Every array in Lemma 2.1 has weighted energy at most $\sum_i i\lambda_i$. Summing the geometric series over all nonsingletons, without the cut constraints, therefore gives

$$a_n \leq \exp\left(\sum_{i=1}^n i\lambda_i\right) \prod_{a < b} \left(1 - \exp\left(-\sum_{k=a}^b \lambda_k\right)\right)^{-1}. \quad (4.1)$$

For example, this follows by multiplying each admissible array's indicator by $\exp(\sum_i i\lambda_i - \sum_{a < b} m_{[a,b]} \sum_{k=a}^b \lambda_k) \geq 1$.

Because λ_i decreases, an interval of length ℓ ending at j has energy at least $\ell\lambda_j$. Grouping by j , and extending lengths $2 \leq \ell \leq j$ to all $\ell \geq 2$, yields

$$\log a_n \leq U_n := \sum_{j=1}^n \left[c\sqrt{j} + L(c/\sqrt{j}) + \log(1 - e^{-c/\sqrt{j}}) \right]. \quad (4.2)$$

Here the last term removes the singleton factor, which is the source of an important logarithmic correction.

For $0 < t \leq c$, Taylor's formula with a bounded remainder on this fixed interval gives

$$\log(1 - e^{-t}) = \log t - \frac{t}{2} + O(t^2).$$

Use Lemma 3.1. The exponential remainders summed over $t = c/\sqrt{j}$ are bounded, $\sum j^{-1/2} = O(\sqrt{n})$, and $\sum j^{-1} = O(\log n)$. Thus

$$U_n = 2c \sum_{j=1}^n \sqrt{j} - \frac{3}{4} \log(n!) + n \left(\frac{3}{2} \log c - \frac{1}{2} \log(2\pi) \right) + O(\sqrt{n}). \quad (4.3)$$

The elementary sum estimate $\sum_{j=1}^n \sqrt{j} = \frac{2}{3}n^{3/2} + O(\sqrt{n})$ and Stirling's formula now give (1.3), including the stated B_U .

Remark 4.1 ([write], 5 October 2026: B_U is not the linear term). The constant $B_U = 0.2043\dots$ in (1.3) bounds from above; it is not a coefficient of n in $\log a_n$, and small cases suggest that the actual linear behaviour lies far below it. The intake evaluated U_n of (4.2) with the exact identity (3.7) (40-digit floating point, `mpmath`; not interval arithmetic) and compared it with the exact $\log a_n$ of the first 26 terms of the OEIS b-file [1]. The inequality $\log a_n \leq U_n$ holds for every $1 \leq n \leq 26$; the gap $U_n - \log a_n$ is 1.39, 14.96, 31.14 and 41.06 at $n = 1, 10, 20$ and 26. For $9 \leq n \leq 26$,

$$\frac{\log a_n - Cn^{3/2} + \frac{3}{4}n \log n}{n} \in [-1.443, -1.421],$$

with the largest value -1.4213 at $n = 10$ and a slow decrease after it (-1.4336 at $n = 20$, -1.4426 at $n = 26$). At these sizes, therefore, $\log a_n \approx Cn^{3/2} - \frac{3}{4}n \log n - 1.4n$, about $1.6n$ below the upper estimate with B_U . Such small n cannot separate a linear term from the allowance $2n \log \log n$ of (1.2) (at $n = 26$, $2 \log \log n \approx 2.36$) or from lower-order terms, so the data are inconclusive about Question 2.

5 An exactly feasible high entropy flow

Fix once and for all

$$M = 1024. \quad (5.1)$$

For $1 \leq a \leq b \leq n$, define

$$q_{[a,b]} = \begin{cases} Q(c(b-a+1)/\sqrt{a}), & a \geq M, \\ 0, & a < M, \end{cases} \quad T_i = \sum_{a \leq i \leq b} q_{[a,b]}. \quad (5.2)$$

There are at most ℓ intervals of length ℓ through cut i , and $a \leq i$ implies $c/\sqrt{a} \geq c/\sqrt{i}$. Hence

$$0 \leq T_i \leq R(c/\sqrt{i}) \leq i. \quad (5.3)$$

Fill the remaining cut load by singletons:

$$f_{[a,b]} = q_{[a,b]} \quad (a < b), \quad f_{[i,i]} = q_{[i,i]} + i - T_i. \quad (5.4)$$

This real array is nonnegative and has cut loads exactly i , hence exactly the required netflow. Define its geometric entropy by $H_g(f) = \sum_{a \leq b} h(f_{[a,b]})$.

Proposition 5.1. *The flow (5.4) satisfies*

$$H_g(f) \geq Cn^{3/2} - \frac{1}{4} \log(n!) - O(n). \quad (5.5)$$

Proof. By monotonicity of h , it is enough to bound $\sum h(q_{[a,b]})$. First temporarily remove the terminal restriction $b \leq n$ and the fixed initial cutoff. A full row at start a contributes, by (3.6),

$$\sum_{\ell \geq 1} E(\ell c/\sqrt{a}) = 2c\sqrt{a} - \frac{1}{4} \log a + O(1).$$

Summing over $1 \leq a \leq n$ gives $Cn^{3/2} - \frac{1}{4} \log(n!) + O(n)$. Removing starts $a < M$ loses only a fixed constant: there are finitely many such infinite rows, each of finite entropy.

The entropy lost on imposing $b \leq n$ is bounded above, before the cutoff, by

$$\begin{aligned} D_n &= \sum_{a=1}^n \sum_{\ell > n-a+1} E(\ell c/\sqrt{a}) \\ &\leq \sum_{\ell \geq 1} \ell E(\ell c/\sqrt{n}) \leq \frac{3\zeta(3)}{c^2} n + \frac{\pi^2}{3c} \sqrt{n}. \end{aligned}$$

The first inequality uses $a \leq n$, the decrease of E , and the fact that at most ℓ start positions contribute for a fixed omitted length. The last inequality is Lemma 3.2. Thus the terminal loss is $O(n)$, not merely an uncontrolled $o(n^{3/2})$. $\square$

Let r_j denote the total outflow at vertex j , $0 \leq j < n$. Feasibility gives

$$r_0 = 1, \quad \text{inflow at } j = r_j - 1 \quad (1 \leq j < n), \quad 1 \leq r_j \leq j+1. \quad (5.6)$$

The last inequality holds because the outgoing edges of j form a subset of cut $j+1$. The inflow at the terminal vertex is n .

Lemma 5.2 (Initial integer throughputs). *For $0 \leq j < M$ with $j < n$, one has $r_j = j+1$.*

Proof. If $i = j+1 < M$, no q interval starts at or before i , so its outgoing row is just the singleton of value i . At $i = M$, every q interval crossing cut M starts at M . Consequently its entire row mass is T_M , and adding the singleton gap $M - T_M$ makes the outgoing total exactly M . This verifies the boundary case $j = M-1$ as well. $\square$

6 Load deficits and bulk throughput control

The main reason the margin penalty is manageable is that most outgoing totals are much smaller than the crude bound $j + 1$.

Lemma 6.1 (Infinite load deficit). *Let T_i^∞ be the load through cut i in the uncut, untruncated array $q_{[a,b]} = Q(c(b-a+1)/\sqrt{a})$, $1 \leq a \leq i \leq b < \infty$. Then*

$$0 \leq i - T_i^\infty = O(\sqrt{i}). \quad (6.1)$$

The same assertion holds after removing the fixed rows $a < M$.

Proof. Put $\lambda = c/\sqrt{i}$. We already have $T_i^\infty \leq R(\lambda) \leq i$, and (3.5) gives

$$R(\lambda) = i - \frac{\sqrt{i}}{2c} + O(1).$$

We show that the inhomogeneity changes the homogeneous sum by at most $O(\sqrt{i})$. For $\ell \leq i/2$, the possible starts of a length- ℓ interval crossing cut i are $a = i - r$, $0 \leq r < \ell$. The mean value theorem gives

$$0 \leq \frac{c}{\sqrt{i-r}} - \lambda \leq K \frac{\lambda r}{i}$$

with an absolute constant K . Write $V(t) = -Q'(t) = e^t/(e^t - 1)^2$; this is positive and decreasing. It follows that

$$0 \leq Q(\ell\lambda) - Q(c\ell/\sqrt{i-r}) \leq K \frac{\lambda}{i} \ell r V(\ell\lambda).$$

After summing in r and ℓ , this part of the discrepancy is at most

$$K \frac{\lambda}{i} \sum_{\ell \geq 1} \ell^3 V(\ell\lambda) = O(\sqrt{i}). \quad (6.2)$$

To check the last bound uniformly, use $V(t) = O(t^{-2})$ on $(0, 1]$ and $V(t) = O(e^{-t})$ on $[1, \infty)$. Splitting at $\ell = 1/\lambda$ gives $\sum \ell^3 V(\ell\lambda) = O(\lambda^{-4})$; when $\lambda > 1$ the bounded interval $1 \leq \lambda \leq c$ is harmless. Since $\lambda = c/\sqrt{i}$, (6.2) is $O(\sqrt{i})$.

For all remaining lengths, the total homogeneous contribution is

$$\sum_{\ell > i/2} \ell Q(\ell\lambda) = O(i^{3/2} e^{-c\sqrt{i}/2}) = O(\sqrt{i}).$$

For sufficiently large i this follows from the geometric-series tail; finitely many i are absorbed in the constant. The inhomogeneous contribution is between zero and this homogeneous tail. Therefore $0 \leq R(\lambda) - T_i^\infty = O(\sqrt{i})$, proving (6.1).

Finally, deleting $a < M$ changes a cut load by at most $\sum_{a < M} \sum_{\ell \geq 1} Q(c\ell/\sqrt{a}) < \infty$. This constant does not depend on i or n . $\square$

Lemma 6.2 (Terminal layer). *Put*

$$B_n = \left\lceil \frac{4}{c} \sqrt{n} \log(n+1) \right\rceil. \quad (6.3)$$

For $M \leq i \leq n - B_n$, the finite, cut-off loads in (5.2) satisfy $i - T_i = O(\sqrt{i})$, uniformly in both i and n .

Proof. An interval crossing cut i and removed by $b > n$ has length greater than $d = n - i$. Its missing load is therefore at most $\sum_{\ell > d} \ell Q(\ell c / \sqrt{i})$. When $dc / \sqrt{i} \geq 1$, compare $Q(t) \leq (1 - e^{-1})^{-1} e^{-t}$ and sum the geometric tail to bound this by

$$K(d\sqrt{i} + i + \sqrt{i}) \exp(-cd / \sqrt{i}). \quad (6.4)$$

If $d \geq B_n$, the exponential is at most $(n + 1)^{-4}$, whereas the prefactor is $O(n^{3/2} + n)$. The missing load is thus $O(1)$. Combine this with Lemma 6.1. If the displayed index range is empty, there is nothing to prove. $\square$

Proposition 6.3 (Uniform bulk bound). *There is an absolute K such that*

$$r_j \leq K\sqrt{j} \log(j + 1) \quad (M \leq j \leq n - B_n - 1). \quad (6.5)$$

The terminal set contains at most $B_n + 1$ vertices and satisfies $r_j \leq j + 1$.

Proof. The full q row at start i has mass

$$\sum_{\ell \geq 1} Q(\ell c / \sqrt{i}) = O(\sqrt{i} \log(i + 1)).$$

For $\ell c / \sqrt{i} \leq 1$ use $Q(t) \leq 1/t$ and the harmonic sum; for larger arguments use an exponential tail. The outgoing row at vertex $j = i - 1$ is a truncated q row plus $i - T_i$. Lemma 6.2 now proves (6.5), after changing the constant to replace i by j . $\square$

A bound needed below is

$$B_n \log(n + 1) = O(\sqrt{n} \log^2(n + 1)) = O(n). \quad (6.6)$$

This uses $\log^2(n + 1) = O(\sqrt{n})$. It makes the entire endpoint layer inexpensive at the present precision, despite its growing size.

7 Independent integer margins from local triangles

For each internal vertex $j = M, \dots, n - 1$, the edge $(j - 1, j + 1)$, equivalently interval $[j, j + 1]$, has initial flow

$$w_j = Q(2c / \sqrt{j}). \quad (7.1)$$

Choose $t_j \in [0, w_j/2]$ and replace the direct two-step edge by the same amount through the intermediate vertex:

$$\begin{aligned} f_{(j-1, j+1)} &\mapsto f_{(j-1, j+1)} - t_j, \\ f_{(j-1, j)} &\mapsto f_{(j-1, j)} + t_j, \\ f_{(j, j+1)} &\mapsto f_{(j, j+1)} + t_j. \end{aligned} \quad (7.2)$$

Here edge subscripts are used, rather than interval subscripts.

Lemma 7.1 (Independence and number of choices). *All the moves (7.2) can be made simultaneously. They preserve nonnegativity and every cut load. The move at j increases r_j by t_j and leaves every other outgoing total unchanged. The choices making every outgoing total integral yield a set $\mathcal{R}_n$ of distinct integer throughput vectors with*

$$\log |\mathcal{R}_n| \geq \frac{1}{2} \log(n!) - O(n). \quad (7.3)$$

Proof. The reduced edges are distinct nonsingletons. All other affected edges are singletons receiving only additions. Thus no choices conflict, and each reduced edge retains at least $w_j/2$. A cut crossed by the removed edge is crossed by exactly one of the two replacement edges, so its load is unchanged. In outgoing row $j - 1$ the decrease and increase cancel; row j increases by t_j ; no other row changes. This proves independence even when consecutive moves share a singleton.

For $j \geq M$, $2c/\sqrt{j} \leq 1$, and $e^x - 1 \leq 2x$ on $[0, 1]$ gives

$$\frac{w_j}{2} \geq \frac{\sqrt{j}}{8c} \geq 2.$$

Every closed real interval of length $w_j/2$ contains at least $\lfloor w_j/2 \rfloor$ integers. Since its length is at least two, this number is at least $\sqrt{j}/(16c)$. Choose $\rho_j \in \mathbb{Z} \cap [r_j, r_j + w_j/2]$ and set $t_j = \rho_j - r_j$. The early throughputs are already integers by Lemma 5.2, and are unchanged. Consequently

$$|\mathcal{R}_n| \geq \prod_{j=M}^{n-1} \frac{\sqrt{j}}{16c}.$$

Taking logarithms gives (7.3); replacing $(n-1)!$ by $n!$ costs only $\frac{1}{2} \log n$. Distinct choices give distinct throughput vectors because each coordinate is independently specified. All statements are for sufficiently large n ; finitely many smaller sizes may be absorbed in constants. $\square$

Lemma 7.2 (Uniform entropy and throughput control). *Every resulting real flow $f^{(\rho)}$, $\rho \in \mathcal{R}_n$, satisfies*

$$H_g(f^{(\rho)}) \geq Cn^{3/2} - \frac{1}{4} \log(n!) - O(n). \quad (7.4)$$

The bulk estimate (6.5) and exact bound $\rho_j \leq j + 1$ continue to hold, uniformly in the chosen vector.

Proof. Only the direct edge in a triangle loses flow. All singleton entropy changes are nonnegative. By (3.3), each direct-edge loss costs at most

$$h(w_j) - h(w_j - t_j) \leq h(w_j) - h(w_j/2) \leq \int_{w_j/2}^{w_j} \frac{du}{u} = \log 2.$$

The total entropy loss is at most $(n - M) \log 2$, proving (7.4) by Proposition 5.1. Also $t_j \leq w_j/2 \leq \sqrt{j}/(4c)$, using $e^x - 1 \geq x$. This preserves the bulk bound. The exact bound follows afresh from preserved cut feasibility. $\square$

8 Capacity counting with the triangular support

We state precisely the external counting theorem used. For a nonnegative integer row vector $\alpha \in \mathbb{Z}_{\geq 0}^m$, column vector $\beta \in \mathbb{Z}_{\geq 0}^p$ of equal total, and cell-bound matrix $K = (k_{ij})$ with $k_{ij} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, let $\text{CT}_K(\alpha, \beta)$ count nonnegative integer matrices with those margins and entries at most k_{ij} . Write

$$P_K(x, y) = \prod_{i,j} \left(\sum_{\ell=0}^{k_{ij}} (x_i y_j)^\ell \right).$$

An infinite upper bound means a convergent geometric series. Capacity is the infimum of $x^{-\alpha} y^{-\beta} P_K(x, y)$ over positive arguments in its convergence domain; divergent geometric factors are assigned $+\infty$. This is the positive-series convention of [3, §5.1], rather than a rational evaluation beyond a pole.

Theorem 8.1 (Brändén–Leake–Pak). *With the preceding hypotheses, put*

$$\lambda_i = \sum_j k_{ij}, \quad \gamma_j = \sum_i k_{ij}, \quad u_i = \min\{\alpha_i, \lambda_i - \alpha_i\}, \quad v_j = \min\{\beta_j, \gamma_j - \beta_j\}.$$

For feasible margins,

$$\text{CT}_K(\alpha, \beta) \geq \left(\prod_{i=2}^m \frac{u_i^{u_i}}{(u_i + 1)^{u_i+1}} \right) \left(\prod_{j=1}^p \frac{v_j^{v_j}}{(v_j + 1)^{v_j+1}} \right) \text{Cap}_{\alpha, \beta}(P_K), \quad (8.1)$$

where the zero-margin factor is interpreted as one. Arbitrary zero patterns and infinite cell bounds are allowed.

The omission of one row factor is part of the source theorem. Only the case with all support row and column sums infinite is used below, so $u_i = \alpha_i$ and $v_j = \beta_j$. Inserting the omitted factor $e^{-h(\alpha_1)} \leq 1$ weakens (8.1) to the convenient symmetric bound

$$\text{CT}_K(\alpha, \beta) \geq \exp \left(- \sum_i h(\alpha_i) - \sum_j h(\beta_j) \right) \text{Cap}_{\alpha, \beta}(P_K). \quad (8.2)$$

No equal-margin hypothesis or complete rectangular support hypothesis is being added.

[write] **Intake comparison with the source of Theorem 8.1, 5 October 2026.** The intake compared Theorem 8.1 with Theorem 2.1 of the arXiv version 2008.05907v3 of [3] (fetched; its SHA-256 equals the fingerprint in `data/sources.json`). The restatement is faithful. The row product does start at $i = 2$, which the source flags as intended (“not a typo”); the truncated margins are $\min\{\alpha_i, \lambda_i - \alpha_i\}$ and $\min\{\beta_j, \gamma_j - \beta_j\}$; the cell bounds are arbitrary elements of $\mathbb{Z}_{\geq 0} \cup \{+\infty\}$, infinite bounds being handled there as limits of large finite ones; and §5.1 there extends capacity to power series with nonnegative coefficients, which is the convention used here. The weakening (8.2) is valid as stated: with infinite row and column support sums $u_i = \alpha_i$ and $v_j = \beta_j$, each factor $x^x/(x+1)^{x+1}$ equals $e^{-h(x)}$, and the added factor $e^{-h(\alpha_1)}$ is at most one. Theorem 8.1 is an external input: ProveIt has not checked its proof.

Fix $\rho \in \mathcal{R}_n$. Use row indices $i = 0, \dots, n-1$ and column indices $j = 1, \dots, n$, allowing precisely the cells $i < j$. The row and column margins of $f^{(\rho)}$ are

$$\alpha = (1, \rho_1, \dots, \rho_{n-1}), \quad \beta = (\rho_1 - 1, \dots, \rho_{n-1} - 1, n). \quad (8.3)$$

Both are nonnegative integer vectors and their totals agree, because $1 + \sum \rho_j = \sum(\rho_j - 1) + n$. Set $k_{ij} = \infty$ when $i < j$ and zero otherwise. Every row and column has an allowed cell, hence infinite support sum. The displayed real flow proves feasibility.

Every integer table with these margins is an integral flow of netflow $(1, \dots, 1, -n)$: the margin difference at internal vertex j is $\rho_j - (\rho_j - 1) = 1$. Conversely every such flow with outgoing vector ρ is one of these tables. Different ρ give disjoint sets, since a flow uniquely determines its outgoing totals.

Lemma 8.2 (Entropy lower bound for capacity). *For any feasible nonnegative real table G with margins (α, β) on this support,*

$$\log \text{Cap}_{\alpha, \beta}(P_K) \geq \sum_{i < j} h(G_{ij}). \quad (8.4)$$

Proof. For $u \geq 0$ and $0 < z < 1$,

$$-\log(1 - z) - u \log z \geq h(u). \quad (8.5)$$

For $u > 0$ minimize the left side at $z = u/(1 + u)$; for $u = 0$ its infimum is zero. Apply (8.5) to $z = x_i y_j$ and $u = G_{ij}$. The margin identities imply $\sum_{i < j} G_{ij} \log(x_i y_j) = \sum_i \alpha_i \log x_i + \sum_j \beta_j \log y_j$. Summing and taking the infimum over the positive convergence domain proves (8.4). This is a one-sided elementary inequality; it assumes neither saddle existence nor strong duality. $\square$

9 Margin summation and completion of the theorem

Lemma 9.1 (Uniform margin penalty). *For every $\rho \in \mathcal{R}_n$, the margins (8.3) satisfy*

$$\sum_i h(\alpha_i) + \sum_j h(\beta_j) \leq \log(n!) + 2n \log \log(n+1) + O(n), \quad (9.1)$$

with the same implied constant for all such vectors.

Proof. Separate the fixed initial segment, the bulk, and the terminal layer. The initial segment has fixed integer margins and costs $O(1)$. The endpoint factors $h(1) + h(n)$ cost $O(\log(n+1))$. In the bulk, Lemma 7.2 and (3.3) give

$$\begin{aligned} h(\rho_j) + h(\rho_j - 1) &\leq 2\log(\rho_j + 1) + 2 \\ &\leq \log j + 2\log \log(j+1) + O(1). \end{aligned}$$

The constants are uniform because $j \geq M$. For the at most $B_n + 1$ terminal vertices, use $\rho_j \leq j + 1$ to bound each pair by $O(\log(n+1))$. Their total is $O(n)$ by (6.6). Summing the bulk logarithms bounds them by $\log(n!)$ and $2n \log \log(n+1)$ respectively. This proves the claim. $\square$

Proof of Theorem 1.1. For each $\rho \in \mathcal{R}_n$, apply (8.2), Lemma 8.2 with $G = f^{(\rho)}$, and Lemmas 7.2 and 9.1. This gives at least

$$\exp\left(Cn^{3/2} - \frac{5}{4}\log(n!) - 2n \log \log(n+1) - O(n)\right) \quad (9.2)$$

integer flows with that margin vector. The estimate is uniform over the family. Summing its disjoint contributions and using (7.3) yields

$$\log a_n \geq Cn^{3/2} - \frac{3}{4}\log(n!) - 2n \log \log(n+1) - O(n). \quad (9.3)$$

Stirling's formula and $n[\log \log(n+1) - \log \log n] = O(1)$ for large n give (1.2). The upper bound is Section 4. Since $\log \log n = o(\log n)$, both the expansion and limit in (1.4) follow. $\square$

Where the growing dimension is controlled. The counting theorem applies for each finite n with explicit margin factors. The feasible entropy, load deficit, boundary estimates, local entropy losses, and margin estimates all have constants independent of n . Therefore the proof does not invoke a fixed-dimensional saddle theorem in a growing dimension, nor an unproved local central limit approximation.

Remark 9.2 ([write], 5 October 2026: the margin penalty is sharp for this family). The loss $2n \log \log n$ in (1.2) is not slack in the estimate of Lemma 9.1: for all sufficiently large n and every $\rho \in \mathcal{R}_n$,

$$\sum_i h(\alpha_i) + \sum_j h(\beta_j) \geq \log(n!) + 2n \log \log n - O(n),$$

with an absolute implied constant. Hence, with (8.1) or (8.2) applied to these margins, the loss can be recovered only by a larger capacity term (an entropy gain of order $n \log \log n$ over $H_g(f^{(\rho)})$), a larger or different family of margin vectors, or a counting inequality with smaller margin factors; see Question 1. (Amended 5 October 2026 after the independent check recorded at the end of Section 12.1: the last paragraph of the proof, which shows that (7.4) and (7.3) are sharp to $O(n)$ and so justifies “only by”, was added, and the second alternative first read “a larger family of margin vectors”).

Proof ([write]). Let $M \leq j \leq n - B_n - 1$ and $a = j + 1$, so $M < a \leq n - B_n$. The outgoing row of vertex j consists of the intervals $[a, b]$, $a \leq b \leq n$; in f it carries $q_{[a,b]}$ plus the nonnegative singleton gap $a - T_a$, and the triangle moves only raise its total, to $\rho_j \geq r_j$. Put $\tau = c/\sqrt{a} \leq c/\sqrt{M} < 1$ and $k = \lfloor 1/\tau \rfloor$. Since $n - a + 1 > B_n \geq (4/c)\sqrt{n} \log(n+1) \geq \sqrt{a}/c \geq k$, every length $\ell \leq k$ is present, and $Q(x) = \frac{1}{2} \coth \frac{x}{2} - \frac{1}{2} \geq \frac{1}{x} - \frac{1}{2}$ gives

$$\rho_j \geq \sum_{\ell=1}^k Q(\ell\tau) \geq \frac{1}{\tau} \log(k+1) - \frac{k}{2} \geq \frac{\sqrt{a}}{c} \left(\frac{1}{2} \log a - \log c - \frac{1}{2} \right).$$

For $a \geq M = 1024$ the bracket is at least $0.39 \log a$, because $(\log c + \frac{1}{2})/\log 1024 < 0.11$. By (3.3), $h(\rho_j) \geq \log(\rho_j + 1) \geq \log \rho_j$ and $h(\rho_j - 1) \geq \log \rho_j$, so

$$h(\rho_j) + h(\rho_j - 1) \geq 2 \log \rho_j \geq \log a + 2 \log \log a - \kappa, \quad \kappa = 2 \log c + 2 \log \frac{1}{0.39}.$$

All other terms of the penalty are nonnegative. Summing over the bulk, $\sum_{a=M+1}^{n-B_n} \log a \geq \log(n!) - (M + B_n) \log n = \log(n!) - O(n)$ by (6.6). For the double logarithms keep only $a \geq \sqrt{n}$, where $\log \log a \geq \log \log n - \log 2$ (the other terms are positive once $a \geq 3$): there are at least $n - B_n - \sqrt{n}$ such a for large n , and $(B_n + \sqrt{n}) \log \log n = O(n)$, so $2 \sum \log \log a \geq 2n \log \log n - O(n)$. Finally $\kappa n = O(n)$.

The estimates (7.4) and (7.3) are also sharp to $O(n)$, so the loss is not hidden in them either. First, ρ_j is chosen only for $M \leq j \leq n - 1$, among the integers in $[r_j, r_j + w_j/2]$, of which there are at most $w_j/2 + 1 \leq \sqrt{j}/(4c) + 1$ (by $e^x - 1 \geq x$); so $\log |\mathcal{R}_n| \leq \frac{1}{2} \log(n!) + O(n)$. Second, the triangle moves never raise a nonsingleton, so $f_{[a,b]}^{(\rho)} \leq q_{[a,b]} \leq Q(c(b-a+1)/\sqrt{a})$ for $a < b$, and (3.6) gives

$$\sum_{a < b} h(f_{[a,b]}^{(\rho)}) \leq \sum_{a=1}^n \sum_{\ell \geq 2} E(\ell c/\sqrt{a}) = Cn^{3/2} - \frac{1}{4} \log(n!) - \sum_{a=1}^n E(c/\sqrt{a}) + O(n),$$

where $E(c/\sqrt{a}) \geq \log(1 + Q(c/\sqrt{a})) \geq \frac{1}{2} \log a - \log c$, since $1 + Q(t) = (1 - e^{-t})^{-1} \geq 1/t$; the nonsingletons therefore contribute at most $Cn^{3/2} - \frac{3}{4} \log(n!) + O(n)$. Every singleton satisfies $f_{[i,i]}^{(\rho)} \leq i$, its cut load; for $M \leq i \leq n - B_n$, moreover, $f_{[i,i]}^{(\rho)} = Q(c/\sqrt{i}) + (i - T_i) + t_{i-1} + t_i = O(\sqrt{i})$ (with $t_{M-1} = 0$) by $Q(t) \leq 1/t$, Lemma 6.2 and $t_j \leq \sqrt{j}/(4c)$ (proof of Lemma 7.2). With (3.3) and (6.6), the singletons contribute at most $\frac{1}{2} \log(n!) + O(n)$, and so $H_g(f^{(\rho)}) = Cn^{3/2} - \frac{1}{4} \log(n!) + O(n)$. Thus the number of margin vectors, the entropy estimate of the capacity and the margin factors used in the proof of Theorem 1.1 are each correct to within a factor $e^{O(n)}$, which is what the assertion after the display needs. $\square$

10 The integer threshold inverse

Define

$$N(y) = \min\{n \geq 0 : a_n \geq y\}, \quad y > 1. \quad (10.1)$$

The sequence is nondecreasing: retain an interval array of size n , put all new intervals ending at $n+1$ equal to zero except the singleton $[n+1, n+1]$ of value $n+1$, and obtain an injective map to size $n+1$. The leading theorem implies $a_n \rightarrow \infty$, so the minimum exists.

Theorem 10.1. *Let $L = \log y$ and $x = (L/C)^{2/3}$. Then*

$$N(y) = x + \frac{1}{2C} \sqrt{x} \log x + O(\sqrt{x} \log \log x) \quad (10.2)$$

$$= \left(\frac{L}{C}\right)^{2/3} + \frac{L^{1/3}}{3C^{4/3}} \log \frac{L}{C} + O(L^{1/3} \log \log L). \quad (10.3)$$

The error is asymptotic with an unspecified constant, not a finite certified numerical error bar or an $O(1)$ inverse.

Proof. Put $b_n = \log a_n$ and $G(t) = Ct^{3/2} - \frac{3}{4}t \log t$ for $t > 0$. Theorem 1.1 gives

$$b_n = G(n) + O(n \log \log n). \quad (10.4)$$

By the definition of the minimum, the exact integer inequalities are

$$b_{N(y)-1} < L \leq b_{N(y)}. \quad (10.5)$$

They do not assume any continuous interpolation of a_n .

First $N(y) \rightarrow \infty$, and $b_n \sim Cn^{3/2}$ in (10.5) gives $N(y) \sim x$. Write $N = N(y)$. Applying (10.4) at both N and $N - 1$ yields

$$G(N - 1) \leq L + O(x \log \log x), \quad G(N) \geq L - O(x \log \log x).$$

The mean value theorem gives $G(N) - G(N - 1) = O(\sqrt{x} + \log x) = O(\sqrt{x})$, since $N \sim x$. Hence the two inequalities together give the two-sided estimate

$$G(N) = L + O(x \log \log x). \quad (10.6)$$

As $L = Cx^{3/2}$, equation (10.6) first implies $C(N^{3/2} - x^{3/2}) = O(x \log x)$. Another mean value estimate on a fixed relative neighborhood of x therefore gives $\delta := N - x = O(\sqrt{x} \log x)$. Taylor expansion with this bound now gives

$$\begin{aligned} G(x + \delta) - L &= \frac{3C}{2}\sqrt{x}\delta - \frac{3}{4}x \log x \\ &\quad + O\left(\frac{\delta^2}{\sqrt{x}} + |\delta| \log x\right) \\ &= \frac{3C}{2}\sqrt{x}\delta - \frac{3}{4}x \log x + O(\sqrt{x} \log^2 x). \end{aligned}$$

Because $\sqrt{x} \log^2 x = o(x \log \log x)$, comparison with (10.6) proves (10.2). Substitution of $x = (L/C)^{2/3}$ proves (10.3).

For an explicit asymptotic integer bracket, let $z = x + (2C)^{-1}\sqrt{x} \log x$ and $s = \sqrt{x} \log \log x$. There is a constant $D > 0$ such that, for all sufficiently large y ,

$$\lfloor z - Ds \rfloor < N(y) \leq \lceil z + Ds \rceil. \quad (10.7)$$

Indeed, at $m_- = \lfloor z - Ds \rfloor$ and $m_+ = \lceil z + Ds \rceil$ the Taylor expansion gives

$$G(m_{\pm}) - L = \pm \frac{3C}{2}Dx \log \log x + o(x \log \log x).$$

Choose D larger than the constant needed to dominate the error in (10.4). Then $b_{m_-} < L \leq b_{m_+}$, and monotonicity gives (10.7). Rounding perturbs the smooth estimate by only $O(\sqrt{x})$ in logarithmic count, already negligible here. This last argument verifies the integer bracketing directly rather than assuming that rounding a continuous inverse is valid. $\square$

[write] On Theorem 10.1, 5 October 2026. The leading inverse $N(y) \sim x$ follows from Corollary 37 of [2] and is not claimed as new; the second term $(2C)^{-1}\sqrt{x} \log x$ and the bracket (10.7) are new here and use only Theorem 1.1 and the monotonicity $a_{n+1} \geq a_n$ proved before the theorem. The two forms agree: with $x = (L/C)^{2/3}$ one has $\sqrt{x} = (L/C)^{1/3}$ and $\log x = \frac{2}{3} \log(L/C)$, so $(2C)^{-1}\sqrt{x} \log x = L^{1/3} \log(L/C)/(3C^{4/3})$. The constant D and the threshold beyond which (10.7) holds are not effective, so the bracket certifies no value of $N(y)$ at a given y ; and since its width $2Ds + 2$ grows like $\sqrt{x} \log \log x$ while the correction term grows like $\sqrt{x} \log x$, the correction dominates the uncertainty only by the factor $\log x / \log \log x$, slowly.

11 Verification scope and reproducibility

The accompanying source package separates three kinds of evidence.

1. The analytic proof above establishes the asymptotic result. Its non-elementary counting input is the stated Brändén–Leake–Pak theorem; the classical eta transformation supplies the one-dimensional partition estimate.
2. Exact integer and rational checks verify finite model identities and local algebra. The checker uses explicit exceptions, so it remains active under Python optimization. Deliberately corrupted inputs must be rejected with nonzero exit status. These computations do not certify a uniform asymptotic estimate or replace any lemma.
3. Floating-point sanity computations illustrate the exponential-profile construction at finite sizes. They are stored separately and are neither interval arithmetic nor exact certificates for transcendental inequalities.

The build instructions, dependency record, checksums, and source provenance are supplied with the editable L^AT_EX source. The PDF is reproducible with the documented deterministic build settings. Source replay and visual page inspection are independent of the mathematical argument.

Finite exact objects. The rational fixtures are finite flows chosen to exercise cut/netflow conversion, singleton completion, simultaneous triangle updates, nonnegativity, integer margin enumeration, and disjointness of outgoing-margin classes. They need not equal the transcendental profile (5.2). The small-size enumerator checks (2.4) against the cut model and the published initial A008608 values. Deliberate perturbations test that the validation rejects inconsistencies rather than silently continuing.

Finite floating observations. The provided sanity program evaluates the genuine profile with $M = 1024$ at $n = 2048$ and $n = 4096$, makes one integer choice at each internal triangle, and recomputes entropy and margins. For example, at $n = 4096$ the observed maximal internal netflow residual is of order 10^{-12} , while the entropy loss is less than $(n - M) \log 2$. These are numerical consistency observations. A floating value of a theoretically valid bound is not itself a directed-rounding certificate.

[write] **Shipped layout and intake reruns, 5 October 2026.** In ProveIt the package is shipped in the repository’s layout, not the delivered one: the programs are in `code/` (`verify_exact.py`, `run_corruptions.py`, `build_fixtures.py`, from the delivered `checks/`; `profile_sanity.py`, from `sanity/`; and `build.sh`), the fixtures, records and source list in `data/`, and the two delivered sub-READMEs as `checks-README.md` and `sanity-README.md`; the report’s `README.md` says how to rerun them on a copy. At placement the intake ran the delivered verifier (21 of 21 files match the manifest), the corruption suite (“2 positive controls and 62 fail-closed corruption rejections”) and the floating sanity program, all on copies of the delivered layout; at the write it reran the exact verifier in both interpreter modes and the sanity program on a copy of the shipped files (Python 3.14.4, NumPy 2.3.5). The exact outputs equal the delivered ones after the carriage returns written by Windows are removed; the sanity run passes, with integer fields and residuals identical and entropies differing in the last three or four digits, as the package says floating results may. Separately, the intake checked (3.7) against the direct series at $t = 0.3, 1$ and c (to 10^{-40}), the upper bound $\log a_n \leq U_n$ for $n \leq 26$ (Remark 4.1), and the recursion (2.4) against $a_1, \dots, a_8$.

12 Further questions

The present result leaves several concrete next steps.

1. **Remove the logarithm of a logarithm loss.** The lower proof pays for $O(\sqrt{j} \log j)$ throughputs via two separate margin factors. Showing $\log a_n = Cn^{3/2} - \frac{3}{4}n \log n + O(n)$ would require recovering the resulting $2n \log \log n$ loss or a sharper counting argument. It does not follow from the estimates proved here.
2. **Identify a linear term.** The upper constant B_U comes from a convenient nonsingleton product bound. There is no proof that it equals an actual next coefficient. Boundary corrections, the optimized entropy action, and the distribution of margin classes may all matter at this scale.
3. **Understand a full coefficient equivalent.** A relative-error equivalent requires control well below the present logarithmic error, including any growing-dimensional lattice and non-Gaussian effects. No determinant approximation, amplitude, or full transseries is asserted.
4. **Sharpen the inverse only after sharpening the forward error.** An $O(n)$ forward remainder would already reduce the inverse uncertainty to $O(\sqrt{x})$ while keeping the same displayed correction. A bounded-error integer inverse would require far finer forward information.
5. **Generalize the cut profile carefully.** The feasibility comparison suggests studying other slowly varying cumulative loads. The netflow, boundary costs, entropy scale, and the availability of independent margin moves would all have to be checked anew; the present proof does not automatically supply a theorem for other profiles.

12.1 Further questions and research

[write] **Added 5 October 2026, batch 103.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. Questions 1–5 are the five items above, sharpened by the write’s Remarks 4.1 and 9.2; Question 6 concerns priority. The intake found no false claim in the source.

1. **An $O(n)$ remainder** (source, item 1). Is $\log a_n = Cn^{3/2} - \frac{3}{4}n \log n + O(n)$? The upper bound (1.3) already has this form; the lower bound loses $2n \log \log n$ through the two margin factors of (8.2), because the bulk throughputs are of order $\sqrt{j} \log j$ (Proposition 6.3). By Remark 9.2 that loss is exact for the family $\mathcal{R}_n$: no sharper estimate of the margin factors removes it. Missing: a capacity bound exceeding $H_g(f^{(\rho)})$ by order $n \log \log n$, a larger or different family of margin vectors (the triangle moves give about $\sqrt{j}$ choices at vertex j , whereas the outgoing totals themselves are of order $\sqrt{j} \log j$; this is a heuristic, not a count), or a counting inequality with smaller margin factors. (Amended 5 October 2026 with Remark 9.2, after the independent check below: “or different” was inserted, and the last paragraph of the remark’s proof now shows that the capacity estimate and the count of margin vectors used are themselves sharp to $O(n)$.)
2. **A linear term** (source, item 2). Does $(\log a_n - Cn^{3/2} + \frac{3}{4}n \log n)/n$ converge, and to what? B_U comes from the product bound (4.1) and is not that limit. For $9 \leq n \leq 26$ the quotient lies in $[-1.443, -1.421]$ and decreases slowly from $n = 10$ on (Remark 4.1); this is far below B_U and too short a range to decide anything. The source names boundary corrections, the optimized entropy action and the distribution of margin classes as possible contributions.

3. **A relative-error equivalent** (source, item 3): an amplitude, a determinant approximation or a full expansion of a_n itself. Missing: control of $\log a_n$ to $o(1)$, including growing-dimensional lattice and non-Gaussian effects; nothing in the present proof approaches this.
4. **A sharper inverse** (source, item 4). Given an $O(n)$ forward remainder (Question 1), the argument of Theorem 10.1 would give $N(y) = x + (2C)^{-1}\sqrt{x} \log x + O(\sqrt{x})$, since $G'(n) \asymp \sqrt{n}$. A bounded-error integer inverse would need a forward error $O(\sqrt{n})$, far beyond the present information.
5. **Other cut profiles** (source, item 5). Do analogues hold for flows with other slowly varying cumulative loads? The netflow, the boundary costs, the entropy scale and the availability of independent margin moves would all have to be checked anew; the present proof supplies no theorem for other profiles.
6. **Priority of the coefficient $-3/4$** (source, “Scope of the refinement”, and the intake). Is the second term new? The source’s check was bounded and the source disclaims novelty certification. The intake did not find the term in [2] (whose “Future questions” do not ask for it), [3], the arXiv version of [6] (bounds only, of orders $e^{n \log n + O(n)}$ below and $e^{O(n^2)}$ above) or the OEIS entry. The journal version of [6] has the same bounds and question as the arXiv version (Theorem 43, Question 4, Remark 5), and Theorem 1.2 of [4] (arXiv v3) is a lower bound of order $n \log^2 n$ only, so neither contains the coefficient. Missing: a search of the Tesler-matrix and Kostant-partition-function literature ([9, 8, 7] and the further references of the OEIS entry) for second-order asymptotics. (Updated 5 October 2026 after the independent check below, which made both comparisons; “Missing:” first began “a comparison with Theorem 1.2 of [4] and with the journal version of [6], which the manuscript does not cite, and a search”.)

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of what this write supplied, made by the intake after the write (17d995b94), read the article in full. It found valid the statement and proof of Remark 9.2, the statements of Section 1.1 that Corollary 37 of [2] answers Pak’s question and that (4.2) alone answers it, and every figure of Remark 4.1; it found no counterexample and no gap in the proof of the remark. Four changes followed, each marked where it stands by a dated note that keeps the first wording. (1) The conclusion of Remark 9.2, that the loss can be recovered “only by” three routes, presupposed two upper bounds that are true but were not proved in the report, $\log |\mathcal{R}_n| \leq \frac{1}{2} \log(n!) + O(n)$ and $H_g(f^{(\rho)}) \leq Cn^{3/2} - \frac{1}{4} \log(n!) + O(n)$; and a different family of margin vectors, with smaller throughputs, need not be a larger one. The check supplied a short proof of both bounds, now the last paragraph of the proof of the remark (checked again step by step when it was inserted), and the remark and Question 1 now say “a larger or different family”. (2) The write cited O’Neill’s results by their arXiv numbers (Theorem 5.6, Question 1.4) against a bibliography entry for the journal version, which numbers them Theorem 43, Question 4 and Remark 5; Section 1.1 and the bibliography now give both numberings, and attribute the $n^{1.6}$ observation, as O’Neill does, to a private communication of J. Pantone. (3) Section 1.1 now records that (4.2) gives $\log a_n \leq C(n+1)^{3/2}$ for every $n \geq 1$, without the eta transformation, and that §7.3 of [4] still names $O(n^2)$ as the best known upper bound. (4) The two comparisons that Question 6 listed as missing were made: the journal version of [6] has only the bounds and question of the arXiv version, and Theorem 1.2 of [4] is the lower bound $\log a_n \geq \frac{n}{4} \log^2 n - O(n \log n)$ (with $a_n \geq (n+1)^{n-1}$ for $n \geq 3000$). Neither contains the coefficient $-3/4$, so the question and Section 1.1 now ask only for the literature search.

The check reread every step of the proof of Remark 9.2 (the row structure, the range $n - a + 1 > B_n \geq \sqrt{a}/c \geq k$, $\tau \leq c/32 < 1$, the chain of inequalities, $(\log c + \frac{1}{2})/\log 1024 = 0.10804 < 0.11$, $\kappa = 2.3809$, and the summation) and tested it with `mpmath` at 30 digits: the identity $Q(x) = \frac{1}{2} \coth \frac{x}{2} - \frac{1}{2}$ holds to $2 \cdot 10^{-29}$, and $2 \log(0.39(\sqrt{a}/c) \log a) = \log a + 2 \log \log a - \kappa$ to $4 \cdot 10^{-30}$; the four members of the chain decrease as claimed at $a = 1025, 2000, 10^4, 10^5, 10^6$ and 10^7 (at $a = 10^6$: $5284.5 \geq 4802.7 \geq 4802.1 \geq 4201.0$); and $(B_n + \sqrt{n}) \log \log n/n$ is 0.116 at $n = 10^6$ and 0.0023 at $n = 10^{10}$. The full truncated row is about 0.52 to 0.55 times $\sqrt{a} \log a/c$, so the constant 0.39 is conservative. For Remark 4.1 its own programs (50 digits; L by the q -Pochhammer product, agreeing with (3.7) to $3 \cdot 10^{-50}$; $a_1, \dots, a_7$ also from its own implementation of (2.4)) reproduced $\log a_n \leq U_n$ for all $1 \leq n \leq 26$ of the OEIS b-file, the gaps 1.39235, 14.9646, 31.1373 and 41.0584, the quotient range from -1.44258 ($n = 26$) to -1.42127 ($n = 10$) with -1.43357 at $n = 20$, a distance from B_U between 1.625 and 1.647, $2 \log \log 26 = 2.362$, and the decimal of B_U . For (3) it checked $c + \pi^2/(6c) = 2c$ to 10^{-30} and, at $n = 26$, $\log a_n = 125.67 \leq U_n = 166.73 \leq 232.74 \leq 239.92$. On the literature it fetched both versions of [6], the journal version of [2] (its SHA-256 equals the fingerprint in `data/sources.json`; Corollary 37 is on p. 17, and its authors call it a drastic improvement of [4, Theorem 1.2]) and arXiv v3 of [4], whose footnote 2 credits the preprint of [2] (communicated 13 June 2024) with $\log a_n \sim C_2 n^{3/2}$, and which presents its Theorem 1.2 only as a first step towards Pak’s question. This was a careful reading with numerical tests, not a formal verification or an external review.

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- [3] P. Brändén, J. Leake and I. Pak, *Lower bounds for contingency tables via Lorentzian polynomials*, *Israel Journal of Mathematics* **253** (2023), 43–90. doi:10.1007/s11856-022-2364-9. Primary version checked: arXiv:2008.05907v3, 13 February 2021, Theorem 2.1 and §5.1.
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- [5] NIST Digital Library of Mathematical Functions, *Dedekind eta modular transformation*, §23.18, Eq. (23.18.5), <https://dlmf.nist.gov/23.18.E5>. Accessed 2 October 2026.
- [6] J. O’Neill, *On the poset and asymptotics of Tesler matrices*, *Electronic Journal of Combinatorics* **25**(2) (2018), P2.4. doi:10.37236/6877. Versions read: the journal version (Theorem 43 on p. 21; Question 4 and Remark 5 on p. 3) and arXiv:1702.00866v2, 21 March 2017 (the same results as Theorem 5.6, Question 1.4 and an unnumbered remark). The remark attributes the $n^{1.6}$ observation to a private communication of J. Pantone to A. H. Morales (2016). [Added by the write; corrected 5 October 2026 after the independent check, which read the journal version. This note first read “Version read by the intake: arXiv:1702.00866v2, Theorem 5.6 and Question 1.4”.]

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